

COMPUTER NETWORKS

Question Bank

UNIT-I

- 1.What is a Computer network? Explain the OSI model with diagram?
- 2.Explain Applications of Computer networks?
- 3.List the advantages and disadvantages of Bus and Ring Topologies?
- 4.Explain Optical Fiber Cable with the advantages and disadvantages?
- 5.Explain the TCP/IP model with diagram?
- 6.Explain about Coaxial cable with the advantages and disadvantages?
- 7.What is Network Topology? Explain different types of Network Topologies?
- 8.List the differences between LAN,MAN,WAN?
- 9.Explain different types of Twisted Pair Cable?
- 10.Explain different types of Computer Networks?

UNIT-II

- 1.Explain Stop and Wait protocol?
- 2.Explain GO-BACK-N sliding window protocol with diagram?
- 3.Mention different types of Error detection methods ,Explain any two methods with example?
- 4.Explain Selective Repeat sliding window protocol with diagram?
- 5.Explain Stop and Wait ARQ protocol with diagram?
- 6.Explain the services provided by Data link layer?
- 7.Explain the HDLC protocol?
- 8.Explain the HDLC protocol information frame?
- 9.Explain the HDLC protocol supervisory frame?
- 10.Explain Point to point protocol(PPP)?

12. What is Cyclic Redundancy Check? Calculate CRC code using CRC12 for the following data: 100100, by using the following generator polynomial: $x^{12} + x^{11} + x^3 + x^2 + x + 1$?

13. Explain how Hamming code can be used for Error detection in computer networks? Calculate 7 bit Hamming code for the following data: 1011?

**14. What is Checksum? Calculate Checksum for the following Data?
Data: 10110011, 10101011, 01011010, 11010101**

UNIT-III

1. Discuss about CSMA/CA protocol and CSMA/CD protocol?

2. Explain different types of Controlled Access Protocols?

3. Explain pure ALOHA and Slotted ALOHA with diagrams?

4. Explain Reservation and Polling Controlled Access protocols?

5. Explain 1-Persistent CSMA and P-persistent CSMA protocols?

6. Explain Non persistent and O-persistent CSMA protocols?

7. Explain Token Passing Controlled Access protocol?